

OVERVIEW OF THE PROCESSING

Controller(s)

Field	Value
Controller	Operator of lots, housing, real-estate allocation, or property-interest deployment
DPO or similar function	Controller DPO or privacy contact
Repository evidence	Consumers/YARD_CASA; lotsClient.js; patent/mainhub/dpia/evidence_index.m

Name of the Processing

Internal name: Shyware Lots / Real Estate Allocation DPIA

SYSTEMATIC DESCRIPTION OF THE PROCESSING

High-level Description of the Processing

Processed Personal Data

#	Personal Data Item	Data Type / Data Subject Category	Sensitivity
1	Applicant/account commitment	Buyer, renter, applicant, or property participant commitment	Medium
2	Eligibility/allocation data	Lottery, bid, queue, preference, or qualification record	High wh
3	Property-interest record	Parcel, lot, unit, contract, redemption, or allocation metadata	Financia
4	Identity/KYC data	Applicant or participant verification	High

Purposes of the Processing Support property allocation, applicant eligibility, auditability, fraud prevention, and dispute handling.

ANALYSIS OF THE PROCESSING

Lawfulness and Minimisation

Customer controllers must assess housing, fair-lending/fair-housing, consumer-protection, and property-law duties. Eligibility criteria and allocation outcomes can materially affect individuals and should be explainable and contestable.

Mainhub Patent Alignment

The Mainhub patent architecture can reduce canonical-state linkage between applicant commitments and allocation records. It does not address whether eligibility criteria are fair, whether allocation logic creates discrimination, whether pricing/allocation coordination creates antitrust risk, or whether applicants receive adequate explanation and appeal. Those remain controller and domain DPIA issues.

Post-Close Buyer Anonymity — Batch-Queue Settlement A naïve sealed-bid auction with `bid_visibility: sealed_until_close` protects bid-to-bidder linkage only until period close. At settlement, the winning bidder’s identity commitment appears as the `recipientCommitment` of a shywire transfer. Even if no canonical join key exists, an observer who sees exactly one settlement fire immediately after lot X closes can infer the buyer by timing — regardless of whether the record contents are readable.

Sealing the settlement records (payload or identity attribute) addresses content visibility, which is already a structural property of the two-list invariant. It does not address timing correlation: a sealed blob that fires 1:1 with a lot close is still a timing oracle. The correct fix is the same pattern used in `shychat-v1: delivery_model: dispatch_queue` queues messages and batch-flushes them, so multiple messages commit in the same block and per-message timing correlation is broken. Applying the same logic to lot settlements: `settlement_dispatch: batch_queue` queues settlement transfers and flushes them in batches, so multiple lot settlements commit in the same block window. An observer who sees lot X close cannot isolate which of the k settlements in the next batch corresponds to lot X.

Configuration	Effect
<code>bid_visibility: sealed_until_award</code>	Bids remain sealed until award is committed, not merely
<code>settlement_dispatch: batch_queue</code>	Settlement transfers queued and batch-flushed; $k > 1$ sett
<code>bidder_identity_mode: anonymous_commitment</code>	Identity never appears as a raw identifier in canonical sta

Batch size as a privacy parameter. Larger batches increase anonymity (higher k) at the cost of settlement latency. The operator configures the flush interval or minimum batch count; smaller auctions with low transaction volume should enforce a minimum batch size of $k \geq 3$ to prevent degenerate single-settlement batches. This is the same design tradeoff as the dispatch-queue pattern in `shychat-v1`.

Authority Partition This is a count-match embodiment. Four structurally separated parties are required:

Authority	Role	
Auth (Cognito)	Account layer; always required; gates access to IDV/KYC stage	C

Authority	Role	
IDV/KYC (Didit)	Verifies user identity on authenticated users	S
Lot administrator	Determines domain eligibility; registered independently at scoping-identifier creation	C
Reconciling authority	Off-chain linkage store + sealer keys	I

Dual co-authorization enforcement (Claim 2): The validation layer must reject single-signature rescission transactions. **VERIFIED (run 26369476939):** *DUAL_COSIG section passes 379/379 — unauthenticated + valid token → 401; authenticated + no token → 400; both co-signatures → 200.*

CONSIDERATIONS ON NECESSITY AND PROPORTIONALITY

The lots/real estate pattern may be proportionate only where transparent eligibility, allocation audit, fraud prevention, and dispute handling are necessary. Canonical unlinkability does not justify opaque allocation rules or hidden adverse action. Production should wait until eligibility criteria, allocation logic, protected-class proxy review, retention, explanation, and appeal workflows are fixed.

RISK ASSESSMENT AND MANAGEMENT

#	Risk	Required Follow-up
1	Unfair allocation or discrimination	Review eligibility criteria, protected-class proxies, and auditability
2	Property/financial profiling	Minimize public linkability of applicant and property interest
3	Inaccurate eligibility	Define correction, dispute, and appeal workflow
4	Over-retention	Define retention for unsuccessful applicants and expired allocations

DPIA Audit Findings

This DPIA is the audit record for lots and real estate allocation deployments. Before production, the DPIA must identify the provider partition across applicant identity or eligibility provider, allocation operator, protected linkage store, reconciling authority, canonical commit authority, payment/escrow provider if any, and telemetry/support providers. Canonical-state unlinkability does not reduce housing, property, fairness, anti-discrimination, explanation, or appeal obligations. The software evidence for the core kernel may be complete while eligibility criteria, allocation logic, notice, retention, appeal, and adverse-action controls still need to be fixed for this domain.

Operator Compliance Instructions For the non-code governance residuals identified in this DPIA, the operator is instructed by the **Shyware SDK Compliance Guide** (patent/mainhub/dpia/source/COM published at <https://shyware.fyi/legal>). Key operator obligations:

- **Art. 6 lawfulness basis** — Identify and document the legal basis for each processing purpose before production.
- **Art. 28 sub-processor DPAs** — Execute signed DPAs with all subprocessors listed in the generated Records of Processing (`node scripts/generate-rop.mjs -config shyconfig.json`).
- **Art. 30 Records of Processing** — Generate from shyconfig using `generate-rop.mjs`; complete all `REQUIRED_OPERATOR_INPUT` fields; maintain under Art. 30(1).
- **Storage limitation (Art. 5(1)(e))** — Add `retention` block to `shyconfig.json`; configure and operationally enforce the `deletion_cron` schedule.
- **Three-authority key independence** — Verify at scoping-identifier creation that `eligibility_authority_key` $\neq$ `reconciling_authority_key` $\neq$ `auth_key`.
- **Quarterly independent audit** — Engage an independent auditor to verify three-authority operational separation before production.

Residual Risk Assessment

The residual scores below assume the core Shyware twin-kernel controls are implemented and evidence-linked. The software evidence for the core kernel may be complete while housing/property eligibility, fairness, explanation, appeal, payment/escrow, or applicant-data choices still need to be fixed for this domain. These scores describe residual domain risk after validation; they do not say the kernel tests were missing.

Risk	Additional Measures Applied
Eligibility or allocation discrimination	Criteria review, fairness testing, notice, and appeal process
Applicant identity or financial exposure	Provider review, minimization, retention limits, and access controls
Adverse-action opacity	Explanation, human review, and appeal workflow
Payment/escrow linkage	Payment provider review, tokenization, and role partitioning
Provider consolidation	Provider matrix for eligibility, allocation, payment/escrow, linkage, re

INVOLVEMENT OF INTERESTED PARTIES

Privacy counsel, housing/property counsel, fairness/compliance stakeholders, security, and customer DPO teams should review this DPIA before production. Data subject or representative consultation should be considered where housing, lending, public allocation, or materially adverse applicant outcomes may occur.

CONCLUSION AND DECISION

Draft decision: Separate DPIA required before property or housing allocation launch.